EmoCNN-4: A Deep Learning Framework for Real-Time Student Engagement Analysis

Mrs. P. Reshma[1]
Assistant Professor, Department of Computer Science and Engineering
Seshadri Rao Gudlavalleru Engineering College
Gudlavalleru, Andhra Pradesh, India
reshmapothuri@gmail.com

Sammangi Lohitha[2]
Department of Computer Science and Engineering
Seshadri Rao Gudlavalleru Engineering College
Gudlavalleru, Andhra Pradesh, India
lohithasammangi12@gmail.com

Veerla Sai Krishna[3]
Department of Computer Science and Engineering
Seshadri Rao Gudlavalleru Engineering College
Gudlavalleru, Andhra Pradesh, India
veerlasaikrishna914@gmail.com

Rajulapati Naga Phanindra Kumar[4]
Department of Computer Science and Engineering
Seshadri Rao Gudlavalleru Engineering College
Gudlavalleru, Andhra Pradesh, India
phanirajulapati2005@gmail.com

***Abstract*-** Student engagement is a key determinant of learning effectiveness, yet traditional assessment methods are often subjective and labor-intensive. This paper presents a real-time automated Student Engagement System that integrates facial emotion recognition, eye-tracking–based attentiveness detection, and face validation within a secure web-based analytics platform. A custom convolutional neural network, EmoCNN-4, is trained on four engagement-related classes—happy, neutral, sad, and surprise—selected from the FER2013 dataset, comprising 18,976 images augmented to enhance robustness. EmoCNN-4 achieves a test accuracy of 78.47%, with high precision for engagement-indicative emotions. Haar-cascade–based eye tracking monitors eye presence and gaze consistency to determine attentiveness. The system includes a face validation mechanism that ensures only authenticated users are analyzed, preventing unauthorized individuals from affecting engagement metrics. Live video analysis, session-wise engagement scoring, and emotion distribution visualization are supported, with persistent metrics capturing total, attentive, and distracted durations per session. Administrative dashboards enable performance monitoring and data-driven insights. Real-time deployment demonstrates accurate detection of attentive and distracted states, reliable emotion recognition, and secure user validation, validating the framework's capability to provide interpretable, scalable, and secure engagement analysis for intelligent educational environments.

Keywords— *Student Engagement, Facial Emotion Recognition, EmoCNN-4, FER2013 Dataset, Eye Tracking, Haar Cascade, Face Validation, Deep Learning, Real-Time Video Analysis, Web-Based Learning Analytics*

I. Introduction

The quick development of online and hybrid learning environments has drastically changed how education is provided, but it has also created new difficulties for tracking student participation. In contrast to traditional classrooms, digital learning platforms do not have direct human supervision, which makes it challenging for teachers to evaluate students' levels of participation, emotional stability, and attentiveness in real time. It is commonly acknowledged that student engagement—which includes emotional, cognitive, and behavioural involvement—is a crucial predictor of academic achievement and learning outcomes. As a result, creating intelligent, automated systems that can assess student participation objectively has emerged as a key area of study in educational technology. Robust analysis of facial expressions and visual signals has been made possible by recent developments in computer vision and deep learning, offering important insights into learners' emotional states. Since motivation and engagement levels are strongly correlated with emotions including happiness, neutrality, sadness, and surprise, facial emotion recognition (FER) has become a popular method for understanding student affect during learning sessions. However, since a student may display neutral emotions when distracted, emotion detection alone could not adequately capture attentiveness. By detecting gaze present and eye visibility, which are potent markers of focus and distraction, visual attention analysis by eye tracking provides complementary information to overcome this constraint. Inspired by these difficulties, this work suggests an all-inclusive Student Engagement System that integrates eye-tracking-based attentiveness detection, real-time web deployment, and deep learning-based face emotion recognition. Haar-cascade-based eye tracking is utilized to identify attentive behaviour, and a custom convolutional neural network called EmoCNN-4 is trained on certain classes from the FER2013 dataset to identify engagement-related emotional states. The solution is implemented as an interactive web application with session-wise engagement metrics, live video analysis, secure user authentication, and administrator dashboards for thorough monitoring. The suggested approach seeks to offer a scalable, impartial, and useful way to improve student engagement analysis in contemporary educational settings by combining emotion analysis, attention estimation, and real-time reporting into a single framework.

II. Literature Reveiw

D'Mello and Graesser investigated the role of affective states such as boredom, confusion, and engagement in learning environments and demonstrated that emotional cues are strong indicators of student engagement and learning outcomes [1]. Their work highlighted the importance of real-time emotion analysis in intelligent tutoring systems. Whitehill et al. proposed an automated approach for measuring student engagement using facial expression analysis and head pose estimation, showing that visual features can effectively predict engagement levels in classroom settings [2]. Mollahosseini et al. introduced a deep convolutional neural network for facial emotion recognition trained on large-scale datasets, including FER2013, achieving robust performance across multiple emotion classes [3]. Their work established CNN-based architectures as a standard for emotion classification tasks. Goodfellow et al. further validated the FER2013 dataset by proposing a deep learning model capable of handling real-world variations such as illumination and pose, making it suitable for practical emotion recognition systems [4]. To complement emotion recognition, eye-tracking and gaze-based attention analysis have been widely explored. Kapoor and Picard presented an affect-aware learning framework that combines visual attention cues with emotional signals to estimate learner engagement [5]. Similarly, Bosch et al. employed facial action units and gaze features to detect student disengagement in real-time learning environments, demonstrating that combining attention cues with emotion recognition improves engagement prediction accuracy [6]. Oztel and Kazanci applied deep transfer learning to facial emotion recognition, showing improved generalization across different datasets, which is relevant for robust engagement detection [7]. Riaz et al. explored shallow CNNs with data augmentation for emotion recognition, demonstrating techniques to enhance model performance with limited data [8]. Pham et al. proposed residual masking networks for facial emotion recognition, improving feature extraction for more accurate emotion classification [9]. Kim and Lee fine-tuned VGGNet for facial emotion recognition, highlighting the effectiveness of pre-trained CNN models for FER tasks [10]. Zahara et al. demonstrated FER-2013-based micro-expression recognition using CNNs, which supports detecting subtle facial cues in engagement analysis [11]. Minaee and Abdolrashidi introduced attentional CNNs for FER, showing that focusing on key facial regions improves emotion prediction accuracy [12]. Khanzada et al. improved deep learning architectures for real-world FER applications, which informs designing

practical engagement systems [13]. Selvaraju et al. developed Grad-CAM for visual explanations of CNN predictions, which can be used to interpret emotion recognition results [14]. Kusuma et al. fine-tuned VGG-16 on FER-2013, achieving higher accuracy, demonstrating that pre-trained networks can accelerate development of reliable FER models [15]. Markapudi et al. proposed a content-based video recommendation system using multilayer neural networks, demonstrating how deep learning can predict user behavior from video streams, which can be adapted for analyzing student attention [16]. Indira et al. developed a multi-perspective CNN for ECG classification, showing that integrating multiple feature perspectives improves classification robustness, an idea that can be applied to combining facial emotion and eye-tracking features for engagement analysis [17].

III. Dataset overview

The facial emotion recognition component of the proposed Student Engagement System is trained and evaluated using the FER2013 dataset, a widely used benchmark dataset for emotion classification tasks. FER2013 consists of grayscale facial images of size 48 × 48 pixels, captured under diverse lighting conditions, facial poses, and expression intensities, making it suitable for real-world emotion analysis. In this work, four emotion classes closely related to student engagement were selected: happy, neutral, sad, and surprise. The dataset includes 5,196 images of happy, 4,965 images of neutral, 4,830 images of sad, and 3,946 images of surprise, along with a small set of augmented surprise samples to improve class representation. All images were preprocessed through grayscale normalization, resizing, and pixel scaling before being used for training and evaluation. The dataset was split into training, validation, and testing subsets to ensure unbiased performance assessment. The diversity and variability of FER2013 enable the EmoCNN-4 model to generalize effectively to real-time video inputs captured in live learning environments.

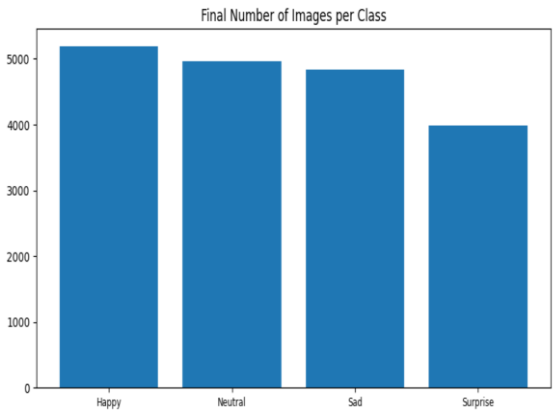

Fig. 1: Distribution of facial images across emotion classes used for model training

IV. Proposed system

The proposed Student Engagement System is designed as an integrated framework that combines facial emotion recognition, eye-tracking–based attentiveness detection, and a web-based real-time analysis platform to objectively evaluate student engagement during learning sessions. The system operates in three major stages: data acquisition, engagement analysis, and reporting & visualization. Live video input is captured through a user's webcam after secure authentication via the web interface. Each login session is uniquely identified, enabling personalized and session-wise engagement tracking and report generation. In the engagement analysis stage, facial frames are extracted from the live video stream and preprocessed through face detection, grayscale conversion, normalization, and resizing to 48 × 48 pixels. These processed frames are then passed to the custom EmoCNN-4 model, which classifies facial expressions into four engagement-related emotion classes: happy, neutral, sad, and surprise. The system uses something called Haar-cascade for detecting eyes, which helps keep track of whether eyes are there and looking steadily, so it can figure out if the person is paying attention or getting distracted when eyes wander off or disappear from the screen. Then, towards the end, it pulls everything together, like showing results and making reports. Over the whole session, it adds up the emotion stuff, how long attention lasted, and those distraction moments to make a full picture of how engaged the user was. The web app gives live visuals, you know, charts on emotions, timelines for attention, and scores on a dashboard that is supposed to be easy to use. After the session wraps up, it spits out a report automatically, stored so teachers or whoever can check back on trends over time. This setup seems pretty solid for handling student engagement in online classes, keeping it accurate and quick without too much hassle.

A. *EmoCNN-4 Model Description*

The EmoCNN-4 model is a custom-designed convolutional neural network developed specifically for grayscale facial emotion recognition in student engagement analysis. The architecture includes three hierarchical convolutional blocks and fully connected layers for classifying the output. This allows it to efficiently extract low- and high-level feature information from a face. In these hierarchical blocks, there are two convolutional layers that use a 3 by 3 kernel, and they have ReLU activation. The filters start at 32, then go up to 64, and end at 128, kind of building up the detail. That setup lets the layers pick out edges and textures from a face, along with stuff specific to expressions. Sometimes I wonder if the increasing filters make that extraction smoother.The max-pooling layers are used after each hierarchical block to reduce spatial dimensions and subsequent computation. Dropout layers with a probability of 0.25 are used to avoid overfitting. Following feature extraction, the network employs fully connected layers to perform high-level reasoning and emotion classification. The flattened output feature maps are fed to dense layers with the number of neurons set to 256 and 128 units, with addition of batch normalization and a dropout rate of 0.5, a higher value used for better generalization. The final output layer is designed with a total of four neurons using the softmax activation function representing the emotion classes "happy," "neutral," "sad," and "surprise." The EmoCNN-4 network is designed to be efficient on the computation required with 48 × 48 grayscale facial image processing. This is functionally important for the system aiming to monitor student engagement in a real-time setting.

V. Methodology

The proposed methodology presents an end-to-end framework for real-time student engagement analysis by integrating facial emotion recognition, eye-tracking–based attentiveness detection, and a web-based analytics platform. The workflow of the system is illustrated through three primary stages: (i) data acquisition and preprocessing, (ii) engagement inference using EmoCNN-4 and eye tracking, and (iii) engagement scoring, visualization, and reporting.

A. *System Overview*

The system operates on live video streams captured through a user's webcam after secure authentication. Each authenticated session is uniquely identified and tracked throughout the learning duration. Facial cues are analyzed using a custom deep learning model (EmoCNN-4) to estimate emotional engagement, while Haar-cascade–based eye detection is used to assess visual attentiveness. The fusion of emotional and attentional cues enables reliable estimation of overall student engagement in real time.

B. *Data Acquisition and Preprocessing*

Live video input is captured at runtime using a standard webcam interface integrated into the web application. Individual frames are extracted at fixed intervals to balance computational efficiency and temporal continuity. For facial emotion recognition, faces are detected using a frontal face Haar cascade classifier. Detected face regions are preprocessed as follows:

- Conversion to grayscale to match the format in FER2013 dataset
- Resizing to 48 × 48 pixels
- Normalization of pixels to the range [0, 1]
- Reshape to match CNN input tensor

These steps ensure that the training and real-time inference are consistent, suppress irrelevant noise, and reduce illumination variations.

C. EmoCNN-4 Architecture and Training

The EmoCNN-4 model is a custom convolutional neural network designed for efficient and accurate facial emotion recognition in real-time environments. The architecture consists of three convolutional feature extraction blocks followed by fully connected classification layers.

Each convolutional block contains:

- Two Conv2D layers with a 3x3 filter size and ReLU activation functions.
- Batch Normalization to remove learning instabilities
- Max-Pooling for spatial down-s
- Dropout (0.25) to combat overfit

In the layers, the filters start small, like 32, then go up to 64, and then 128 as you go deeper. This way, it builds up from detecting simple edges at first to more complex patterns later on, kind of in steps. It seems like that hierarchical thing makes sense for how learning happens.

After feature extraction, the network applies:

- A Flatten layer to convert feature maps into a one-dimensional vector
- Two dense layers with 256 and 128 neurons, respectively, followed by Batch Normalization and Dropout (0.5)
- A Softmax output layer containing four neurons corresponding to happy, neutral, sad and surprise.

Training wise, it goes with categorical cross-entropy as the loss function, and the Adam optimizer does the work. They ran it for 50 epochs, but included early stopping plus some validation checks, you know, to keep overfitting from happening. I might be missing a detail there, but thats the main idea.

D. Eye-Tracking–Based Attentiveness Detection

To complement emotion recognition, the system incorporates attentiveness analysis using Haar-cascade–based eye detection. This module continuously detects the presence and consistency of eye regions within the facial frame. Attentiveness is inferred based on:

- Continuous detection of both eyes
- Stable eye position across consecutive frames
- Alignment of gaze toward the screen

If eyes are consistently detected and aligned, the user is classified as attentive. Prolonged absence of eye detection or inconsistent gaze indicates distraction. This lightweight approach enables efficient attention monitoring without requiring specialized eye-tracking hardware.

E. Engagement Score Computation

The engagement score is computed by fusing emotional and attentiveness information over time. Emotional states associated with higher engagement (happy and surprise) are assigned higher weights, while neutral and sad states contribute lower engagement values. Attentiveness duration further modulates the engagement score.

For each session:

- Emotion predictions are aggregated across frames
- Attentive and distracted time intervals are calculated
- Normalized engagement scores are calculated for the entire session

The multimodal fusion ensures a more accurate and interpretable engagement estimation compared to an emotion recognition system.

$$E=\alpha E_{emotion}+\beta E_{attention}$$

Where:

- $E_{emotion}$ = weighted emotional score
- $E_{attention}$ = attentiveness ratio
- α,β = weighting coefficients

F. Web-Based Deployment and Analytics

The entire system is implemented as a secure web app, which supports user authentication, real-time video testing, and visualizing functionality. In this, the backend deals with video processing, inference, and storing in the database, while the frontend offers interaction analytics.

Each session generates:

- Emotion distribution charts
- Attentiveness timelines
- Overall engagement scores
- Session-wise engagement reports

Administrative dashboards that track user and session engagement trends provide instructors with data that can inform instructional decisions.

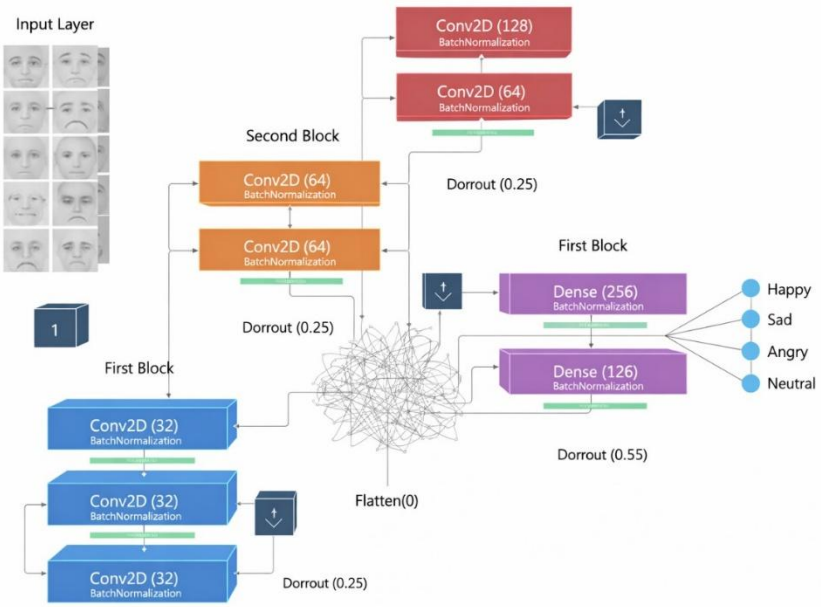

Fig. 3: Proposed CNN Architecture for Facial Emotion Recognition

VI. System Interface

A. *Authentication of the User and Initializing the Session*

The system, as depicted in Fig 4, first starts with a secure user authentication module, where the student logs in using their registered credentials. The authentication mechanism here uniquely identifies each user session in order to enable personalized tracking of engagement. Upon successful login, the system initializes a new session and associates all subsequent emotion and attention analysis results with the authenticated user. This design supports session-wise reporting and historical engagement analysis.

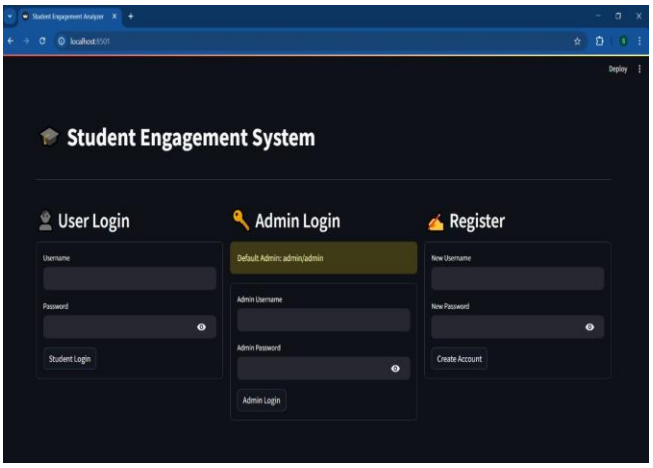

Fig. 4: Login, admin authentication, and user registration interface

B. *Facial Emotion Recognition Module*

The detected facial frames are preprocessed through grayscale conversion, normalization, and resizing before being forwarded to the EmoCNN-4 model. The process of classification of each frame of the video is shown in the Fig 5. The classification of each frame of the video is done based on the emotion, which can be one of the four categories, namely: happy, neural, sad, or surprise. The emotion prediction is shown on the interface in a manner that enables continuous monitoring of the emotion of the learner based on the session.

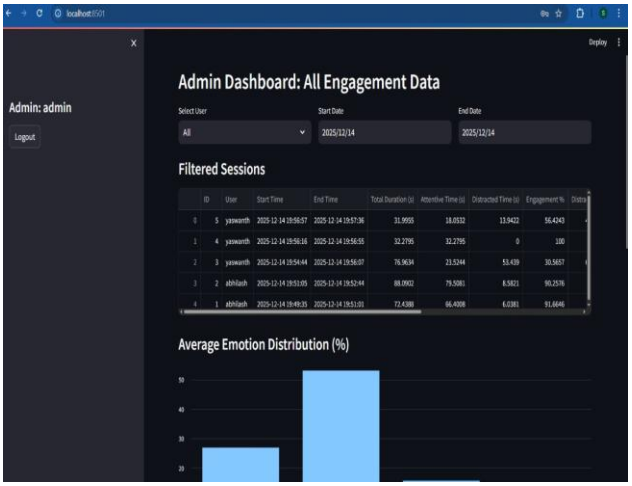

Fig. 5: Administrative dashboard displaying filtered engagement sessions and average emotion distribution

B. *Eye-Tracking–Based Attentiveness Detection*

To complement emotion recognition, the system integrates Haar-cascade–based eye tracking, as shown in Fig 6. This module is responsible for continuously tracking the presence and fixation of the eyes to assess whether the user is paying attention or distracted. Based on the presence and fixation of the eyes on the screen, the user is labeled as focused or inattentive.

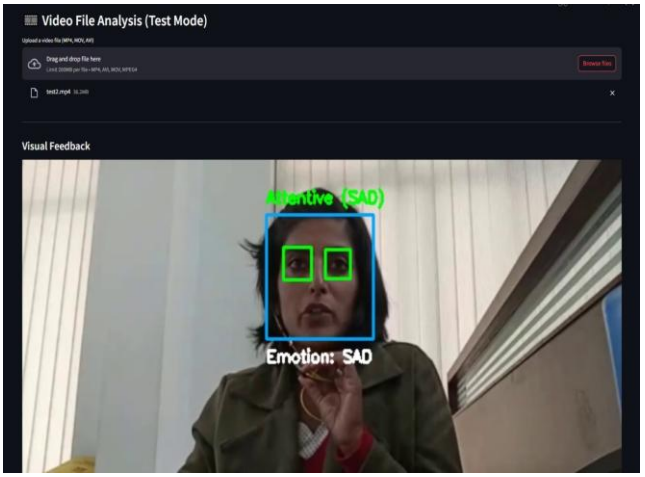

Fig. 6: Emotion and attention detection results during video file analysis, highlighting facial landmarks and predicted emotional state

C. *Real-Time Engagement Analysis and Visualization*

The system fuses emotion recognition results with attentiveness data to compute a real-time engagement score. The web interface dynamically visualizes key metrics, including current emotion, attention status, and engagement indicators. These visual elements allow both users and instructors to intuitively understand engagement patterns during the session, as demonstrated in the attached interface screenshots.

Fig. 7: Average emotion distribution and classification of student sessions into highly engaged and low engaged categories based on engagement thresholds

VII. Results

The proposed Student Engagement System's performance has been assessed by utilizing the FER2013 emotion recognition problem involving four classes of emotion recognition, namely happy, neutral, sad, and surprise. For Emotion-CNN-4, 50 epochs of training were conducted, resulting in a final accuracy of 77.24% on the train set with a validation accuracy of 78.48%. This strongly shows that the model has converged well with little overfitting. The validation loss of 0.5428 is almost equal to the train loss of 0.5812. On the separate test set, it attained a test accuracy of 78.47% with a test loss of 0.5346.

A. *Confusion Matrix Analysis*

A detailed analysis using the classification report further highlights the model's effectiveness across individual emotion classes. The happy class got a good precision of 0.92 and a good recall of 0.85, indicating good recognition of positive signs of engagement. Likewise, the surprise class performed well in terms of having a good precision of 0.88 and a good recall of 0.84, suggesting the ability of the system to detect the facial expressions of attention and interest. The neutral and sad classes performed relatively poorly but steadily, which may be attributed to their subtle expressions and similarity between classes, a recognized issue in facial emotion recognition. The macro average F1 score of 0.78 and weight average F1 score of 0.79 of all classes show relative equality in good classification.

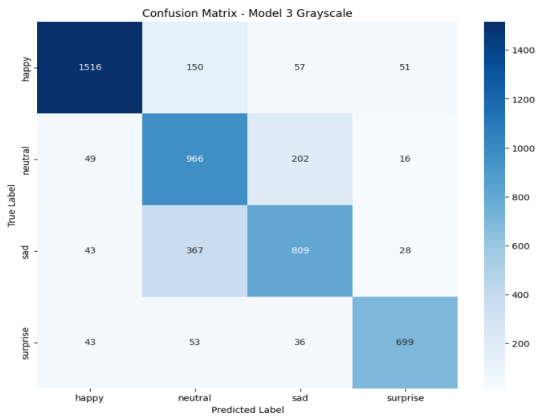

Fig. 8: Confusion Matrix of the Proposed CNN

B. *Quantative Analysis*

The grayscale CNN model demonstrates a steady and consistent improvement in performance over 50 training epochs**,** achieving a final training accuracy of approximately 77.24% and a validation accuracy of around 78.48% as shown in Fig.10**.** The rapid increase in accuracy during the initial 5–10 epochs indicates effective feature learning from grayscale facial inputs, while the gradual convergence afterward reflects stable optimization. The small and stable difference (~1-2%) between the training and validation accuracy typically indicates a good generalization capability and lack of major overfitting problems. The small variations in validation accuracy can be considered normal variations based on changes in facial expressions and illumination conditions but are within acceptable limits. The close matching of the training and validation curves tends to ensure robustness and validate effective learning of discriminative gray-scale feature extraction for this problem.

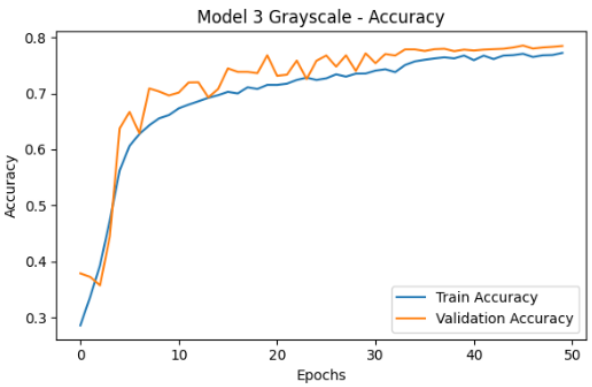

Fig. 9: Training Performance of the Proposed CNN

```
Epoch 43: ReduceLROnPlateau reducing learning rate to 4.0000001899898055e-05.
593/593 [==============================] - 159s 268ms/step - loss: 0.5969 - accuracy: 0.7614 - val_loss: 0.5575 - val_accuracy: 0.7790 - lr: 2.0000e-04
Epoch 44/50
593/593 [==============================] - ETA: 0s - loss: 0.5886 - accuracy: 0.7676
Epoch 44: val_accuracy did not improve from 0.77977
593/593 [==============================] - 162s 273ms/step - loss: 0.5886 - accuracy: 0.7676 - val_loss: 0.5500 - val_accuracy: 0.7798 - lr: 4.0000e-05
Epoch 45/50
593/593 [==============================] - ETA: 0s - loss: 0.5842 - accuracy: 0.7684
Epoch 45: val_accuracy improved from 0.77977 to 0.78210, saving model to .\model_3.keras
593/593 [==============================] - 172s 291ms/step - loss: 0.5842 - accuracy: 0.7684 - val_loss: 0.5471 - val_accuracy: 0.7821 - lr: 4.0000e-05
Epoch 46/50
593/593 [==============================] - ETA: 0s - loss: 0.5843 - accuracy: 0.7709
Epoch 46: val_accuracy improved from 0.78210 to 0.78560, saving model to .\model_3.keras
593/593 [==============================] - 169s 285ms/step - loss: 0.5843 - accuracy: 0.7709 - val_loss: 0.5448 - val_accuracy: 0.7856 - lr: 4.0000e-05
Epoch 47/50
593/593 [==============================] - ETA: 0s - loss: 0.5948 - accuracy: 0.7651
Epoch 47: val_accuracy did not improve from 0.78560
593/593 [==============================] - 179s 301ms/step - loss: 0.5948 - accuracy: 0.7651 - val_loss: 0.5459 - val_accuracy: 0.7802 - lr: 4.0000e-05
Epoch 48/50
593/593 [==============================] - ETA: 0s - loss: 0.5904 - accuracy: 0.7681
Epoch 48: val_accuracy did not improve from 0.78560
593/593 [==============================] - 155s 261ms/step - loss: 0.5904 - accuracy: 0.7681 - val_loss: 0.5470 - val_accuracy: 0.7821 - lr: 4.0000e-05
Epoch 49/50
593/593 [==============================] - ETA: 0s - loss: 0.5868 - accuracy: 0.7687
Epoch 49: val_accuracy did not improve from 0.78560
593/593 [==============================] - 165s 279ms/step - loss: 0.5868 - accuracy: 0.7687 - val_loss: 0.5452 - val_accuracy: 0.7833 - lr: 4.0000e-05
Epoch 50/50
593/593 [==============================] - ETA: 0s - loss: 0.5812 - accuracy: 0.7724
Epoch 50: val_accuracy did not improve from 0.78560
593/593 [==============================] - 173s 292ms/step - loss: 0.5812 - accuracy: 0.7724 - val_loss: 0.5428 - val_accuracy: 0.7848 - lr: 4.0000e-05
```

Fig. 10: Training Log of EmoCNN-4 Showing Learning Rate Scheduling and Convergence Across Final Epochs

C. Comparative Evaluation

The comparative analysis of the newly designed EmoCNN-4 model is presented in Table 1 against several prevailing models in the area of facial emotion recognition on the FER2013 dataset. Some traditional deep learning transfer models like VGG16 and VGGNet obtain a test accuracy of 70.52% and 73.28%, respectively, on the FER2013 dataset, realizing the shortcomings of heavy deep learning models on subtle changes in facial expressions. The eXnet and Residual Masking Network are moderate models that obtain an accuracy of 73.54% and 74.14%, respectively. The Ensemble CNN model, which combines multiple CNN architectures, attains a higher accuracy of 76.82%**,** at the cost of increased computational complexity.

However, the new EmoCNN-4 model shows a better result with a test accuracy of 78.47% on the same problem of classifying emotion into just 4 classes. The reason for this better result is the use of specially designed convolutional blocks with Batch Normalization and Dropout, which improve the stability of the feature representation while preventing it from being prone to overfitting. EmoCNN-4 is entirely different from the ensemble or transfer learning method in the sense that it is highly efficient with a minimal computational approach, ideal for a real-time interaction analysis task. The experiments have proven that task-based optimization of the network architecture can beat even deeper networks with improved generalization performance.

Method / Model	**Technique Used**	**Test Accuracy (%)**
VGG16 (Transfer Learning)	Pretrained VGG16 with fine-tuning	70.52
Residual Masking Network	CNN with residual attention masking	74.14
Ensemble CNN Model	Multi-model ensemble CNN	76.82
Fine-tuned VGGNet	Deep CNN with transfer learning	73.28
Proposed EmoCNN-4	**Custom CNN + BN + Dropout**	**78.47**

Table 1: Performance Comparison of Different Models

Conclusion

This paper presented a comprehensive Student Engagement System that integrates facial emotion recognition, eye-tracking–based attentiveness detection, and a real-time web-based analytics platform to objectively assess learner engagement in digital learning environments. A custom deep learning model, EmoCNN-4, was developed and trained on selected classes from the FER2013 dataset to recognize engagement-related emotional states, achieving a test accuracy of 78.47% with balanced performance across all classes. The experimental results demonstrate that the proposed model effectively captures emotional cues associated with student engagement while maintaining computational efficiency suitable for real-time deployment.In addition to emotion recognition, the incorporation of Haar-cascade–based eye tracking enhances the reliability of engagement assessment by identifying attentive and distracted behaviors. The seamless integration of these components into a secure web application enables live video analysis, session-wise engagement scoring, and automatic report generation for authenticated users. The system's capacity to enable understandable visual representations and a comprehensive engagement report makes it a feasible system to use in education for tracking and optimizing students' engagement in online and combined learning settings. In particular, this proposed system provides a scalable, automatic, and objective system for students' engagement analysis, which incorporates both affective and behavioral measures in a holistic system. The results demonstrate the effectiveness of the EmoCNN-4 model and proposed system architecture.

References

[1] D'Mello, S.; Graesser, A. Dynamics of affective states during complex learning. Learning and Instruction, 2012, 22, 145–157.

[2] Whitehill, J.; Serpell, Z.; Lin, Y.C.; Foster, A.; Movellan, J.R. The faces of engagement: Automatic recognition of student engagement from facial expressions. IEEE Transactions on Affective Computing, 2014, **5**, 86–98.

[3] Mollahosseini, A.; Chan, D.; Mahoor, M.H. Going deeper in facial expression recognition using deep neural networks. IEEE Winter Conference on Applications of Computer Vision (WACV), 2016, 1–10.

[4] Goodfellow, I.; Erhan, D.; Carrier, P.L.; Courville, A.; Mirza, M.; Hamner, B.; Bengio, Y. Challenges in representation learning: A report on three machine learning contests. Neural Information Processing, 2013.

[5] Kapoor, A.; Picard, R.W. Multimodal affect recognition in learning environments. Proceedings of the 13th Annual ACM International Conference on Multimedia, 2005, 677–682.

[6] Bosch, N.; D'Mello, S.; Baker, R.; Ocumpaugh, J.; Shute, V. Automatically detecting student disengagement using facial expressions and eye gaze. Proceedings of the 25th International Joint Conference on Artificial Intelligence (IJCAI), 2016, 152–158.

[7] Oztel, I.; Kazanci, O. Facial emotion recognition using deep transfer learning. *Scientific Reports*, 2024.

[8] Riaz, A.; Bashir, M.; et al. Emotion recognition using shallow CNNs with data augmentation.

[9] Pham, T.; Nguyen, H.; et al. Residual masking networks for facial emotion recognition. *Electronics*, 2023.

[10] Kim, J.; Lee, S. Fine-tuned VGGNet for facial emotion recognition. *Applied Sciences*, 2022.

[11] L. Zahara, P. Musa, E. Prasetyo Wibowo, I. Karim and S. Bahri Musa, "The Facial Emotion Recognition (FER-2013) Dataset for Prediction System of Micro-Expressions Face Using the Convolutional Neural Network (CNN) Algorithm based Raspberry Pi," *2020 Fifth International Conference on Informatics and Computing (ICIC)*, Gorontalo, Indonesia, 2020.

[12] Minaee S., Abdolrashidi A., "Deep-Emotion: Facial Expression Recognition Using Attentional Convolutional Network", 2019

[13] A. Khanzada, C. Bai, and F. T. Celepcikay, *"Facial Expression Recognition with Deep Learning: Improving on the State of the Art and Applying to the Real World,"* Stanford University, CS230: Deep Learning, Project Report, 2019. Available: GitHub CS230 FER Project.

[14] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in 2017 IEEE International Conference on Computer Vision (ICCV), Oct 2017.

[15] G. P. Kusuma, Jonathan, and A. P. Lim "Emotion Recognition on FER-2013 Face Images Using Fine-Tuned VGG-16", Advances in Science, Technology and Engineering Systems Journal, vol. 5, no. 6, pp. 315–322, 2020

[16] Baburao Markapudi, Kavitha Chaduvula, D.N.V.S.L.S. Indira & Meduri V. N. S. S. R. K. Sai Somayajulu, "Content-based video recommendation system (CBVRS): a novel approach to predict videos using multilayer feed forward neural network and Monte Carlo sampling method", published in the journal of Multimedia Tools and Applications (Springer Nature), published online on 11th August 2022, VOL.82, Issue.2, PP:6965–6991. (3 citations)

[17] Devaganugula N.V.S.L.S. Indira , Vyakaranam Sita Maha Lakshmi , Babu Rao Markapudi, Adilakshmi Yannam , Munagala Babu Prasad , Chandanapalli Suresh Babu , Kodepogu Koteswara Rao, " Detection of Cardiac Arrhythmia Using Multi-Perspective Convolutional Neutral Network for ECG Heartbeat Classification", published in the journal of Revue d'Intelligence Artificielle, Vol. 36, No. 4, 31st August, 2022, pp. 629-634 (5 citations)